

DATA ENGINEERING

Assignment 2: SQL Fundamentals & Data Aggregation

Data Manipulation • Scalar Transformations • Grouping & Aggregations

Assignment Goal

Write robust SQL queries to manipulate data (DML), transform attributes using scalar functions, and generate meaningful business insights using aggregate functions, GROUP BY, and HAVING clauses.

Learning Outcomes

- Write INSERT, UPDATE, and DELETE statements to manage data safely.
- Use scalar functions (e.g., string, numeric, date) to clean and transform raw data.
- Summarize datasets using aggregate functions (SUM, COUNT, AVG, MAX, MIN).
- Group data logically and apply post-aggregation filtering using the HAVING clause.

Database Setup Instructions (MANDATORY)

Required Database

For this assignment, you will use the official Microsoft "Northwind" Sample Database. Do NOT create your own tables or assume data types.

- Database Name: Northwind
- Database Type: Relational SQL Server Database
- Size: ~5 MB
- Direct Download Link:

<https://github.com/microsoft/sql-server-samples/tree/master/samples/databases/northwind-pubs>

Installation Steps:

1. Click the link above, right-click on the page, and select "Save As..." to save the file as 'instnwnd.sql'.
2. Open SQL Server Management Studio (SSMS) or Azure Data Studio.
3. Open the downloaded 'instnwnd.sql' file in a new query window.
4. Click "Execute" (or Run). This will automatically create the 'Northwind' database and populate all tables with standard data.
5. Begin your assignment script with: USE Northwind; GO

General Instructions

- Write your queries using standard SQL Server syntax.
- You **MUST** use the exact column and table names as defined in the Northwind database.
- Comment your code briefly to explain the logic behind complex conditions.
- Submit a single .sql file containing all tasks, clearly separated by comments (e.g., -- Task 1.1).

Task 1 - Data Manipulation (DML)

Weight: **20 marks**

1.1 Write an UPDATE query to increase the 'UnitPrice' in the [Products] table by 10% strictly for products where the 'CategoryID' is exactly 8 (Seafood).

1.2 Write an INSERT query to add a new record into the [Shippers] table. Use the exact values: CompanyName = 'FastTrack Delivery', Phone = '(503) 555-1234'. (Do not insert ShipperID, as it is an IDENTITY column).

1.3 Write a safe DELETE query to remove the shipper you just inserted from the [Shippers] table strictly where the 'CompanyName' is exactly 'FastTrack Delivery'.

Task 2 - Scalar Functions & Filtering

Weight: **40 marks**

2.1 Write a SELECT query retrieving the 'ContactName' from the [Customers] table entirely in uppercase aliased as 'FullName'. Additionally, retrieve the 'Phone' column but mask the first 4 characters with '****' using string functions (e.g., STUFF or SUBSTRING).

2.2 Write a SELECT query to calculate the exact number of days between 'OrderDate' and 'ShippedDate' from the [Orders] table using the DATEDIFF() function. Alias this calculated column as 'ProcessingDays'. Filter the results to include ONLY orders where the 'ShipCountry' is exactly 'France'.

2.3 Write a SELECT query to extract only the first name from the 'ContactName' column in the [Customers] table (extract the string from the beginning up to the first space using SUBSTRING and CHARINDEX). Alias this column as 'FirstName'.

Task 3 - Aggregations, GROUP BY, & HAVING

Weight: **40 marks**

3.1 Write a SELECT query to calculate the total net revenue generated per 'ProductID' from the [Order Details] table. The formula for net revenue is: SUM(Quantity * UnitPrice * (1 - Discount)). Alias this calculation as 'TotalRevenue'. Group the results by 'ProductID'.

3.2 Write a SELECT query joining the [Categories] and [Products] tables to generate a report showing the 'CategoryName' and the exact COUNT of products available in each category. Alias the count as 'ProductCount'. Group by 'CategoryName'.

3.3 Write a SELECT query to find the 'CustomerID's of customers in the [Orders] table who have placed more than 15 orders in total. Use GROUP BY and the HAVING clause to apply the filter.

Marking Rubric

Area	Marks	Evaluation focus
Task 1: DML	20	Correct syntax, safe updates/deletes matching the Northwind schema.
Task 2: Scalar Functions	40	Accurate application of string, date, and numeric functions. Proper aliasing.
Task 3: Aggregations	40	Correct logical grouping, accurate aggregates, and proper distinction between HAVING and WHERE.
Total	100	Accurate, efficient, and well-commented SQL scripts conforming to the standard database.

Submission Checklist

- Use the official Northwind database.
- Begin the script with USE Northwind; GO.
- Use exact Northwind table and column names.
- Separate every task clearly in the submitted .sql file.